Venkata Lakshmi Parimala Pasupuleti

AI Platform Engineer · RAG Infrastructure · Agent Harnesses · Document Intelligence · Provenance Tracking · Production AI Systems

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PROFESSIONAL SUMMARY

AI platform engineer with 9+ years of production engineering experience who has already built the platform this role describes - RAG infrastructure over compliance documents, agent harnesses with subagents and tool integrations, and citation-level provenance tracking so non-technical stakeholders can verify every generated output against its source. Leasel is the proof point: a three-surface production AI system I built end-to-end that runs RAG over thousands of complex regulatory documents, routes through an 8-agent LangGraph pipeline with conditional HITL escalation, and enforces 100% citation correctness and groundedness on every output - the document intelligence and review workflow pattern maps directly to investment memo generation and DDQ automation. AgentProbe completes the platform layer: multi-provider RAG infrastructure (Anthropic, OpenAI, Ollama), 15-endpoint FastAPI service, and a CI-integrated regression gate that catches quality degradation before it reaches end users. I use Claude Code as my primary development mode daily - orchestrating parallel engineering tasks, review cycles, and deployment through it. At Entrata, I owned compliance-sensitive production systems for 9+ years (HUD income-limit pipeline, PHI-compliant AWS infrastructure, 99.5% uptime) at the reliability and audit standard financial professionals expect. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Platform & Agent Infrastructure: Python (9yr), Claude Code (daily primary), LangGraph, LangChain, Multi-Agent Orchestration, Agent Harnesses, Subagents, Tool Integrations, HITL, Guardrails, LLM Routing, Anthropic API, OpenAI API, Ollama, TypeScript, JavaScript

Document Intelligence & RAG: RAG Pipelines, ChromaDB, Vector Databases, Document Parsing, OCR, Section-Aware Chunking, Embeddings, Provenance Tracking, Source Attribution, Citation Correctness, Review Workflows, LLM-as-Judge, Hallucination Detection, BERT, spaCy, NLP

Production Infrastructure: FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, PostgreSQL, Redis, RabbitMQ, GitHub Actions CI/CD, Data Ingestion Pipelines, Performance Engineering, Production Observability, PHI/HIPAA Compliance, Data Governance, Agile/Scrum, React.js

PROJECTS

Leasel: Document Intelligence + Agent Harness Platform with Provenance Tracking

Production AI · Document Intelligence · Provenance · RAG

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production AI platform built end-to-end that reads complex regulatory documents, generates structured outputs with full source attribution, and gates every regulated decision behind a human reviewer - the exact pattern for investment memo generation and DDQ automation over financial documents. **Document intelligence layer:** OCR-fallback ingestion, section-aware chunking, TF-IDF RAG retrieval, and 100% citation correctness - every output cites the exact source clause or is explicitly flagged as unsupported; human-review gate on all regulated decisions. **Agent harness layer:** 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional HITL escalation when confidence is below threshold. **Review workflow:** provenance-linked outputs that let non-technical stakeholders verify generated content against source documents. 93.6% precision, 87.9% F1 across 1,240 test cases; zero unsupported-claim rate on the Phase 1 eval set.

AgentProbe: Open-Source RAG Infrastructure & Agent Evaluation Platform

RAG Infrastructure · Agent Harnesses · Production · CI Gate

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production platform I built end-to-end that provides the infrastructure layer for deploying and validating AI agents - multi-provider RAG infrastructure routing across Anthropic, OpenAI, and Ollama via a unified provider interface; 15-endpoint FastAPI service; 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate); CI-integrated regression gate that blocks quality degradation before it reaches end users. Used to evaluate Leasel across 1,240+ compliance test cases. Open-source and deployable in enterprise environments. Zero regressions in production.

AI on FHIR: Compliance-Sensitive Document Intelligence - HIPAA Production

Compliance · APIs · Healthcare · Full-Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack production AI system translating natural-language queries into structured API calls over complex clinical documents - HIPAA-compliant RBAC, authentication, and full data isolation; ~60% reduction in manual retrieval time. Demonstrates both compliance-sensitive document intelligence (FHIR standard) and the stakeholder translation pattern: non-technical users ask questions in plain language, the system returns sourced, structured answers from authoritative records.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - End-to-End Production AI Deployment

- Led end-to-end deployment of a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - from technical discovery through production handoff; translated ambiguous stakeholder requirements into a working system in weeks, serving as sole technical owner through implementation, testing, and delivery. RAG layer cited FAA regulatory corpus so every compliance output was grounded in source regulation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students; translated complex research outputs for academic and non-technical audiences including faculty leadership.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned the compliance-sensitive document pipeline that financial professionals and regulators rely on** - HUD income-limit data ingestion (federal source rows → 180,000 domain records); eligibility, rent, and tax-credit calculations for 100K+ housing units; annual data ingestion with baseline validation, reconciliation, and rollback preparation; zero compliance violations across 7yr of ownership. The production reliability and audit standard this pipeline maintained is the same standard investment professionals expect from a document AI platform.
- **Production infrastructure at enterprise reliability** - AWS S3/EC2/RDS/Glue; Docker (50% scalability improvement, 30% deployment time reduction); RabbitMQ async processing with retry logic and idempotency guarantees; Redis caching; 99.5% uptime across multi-tenant SaaS; 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across 7yr of production operation.
- **Stakeholder translation at scale** - 120+ Agile sprints with product, compliance, and operations stakeholders; built internal debugging tools reducing triage time 40%; 300+ peer code reviews; 20+ customer-facing production modules shipped end-to-end; leveraged Claude Code to accelerate engineering delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data ingestion pipeline for 1.6M housing records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017